years ago was in absolute terms averaged at Rs 200 per month. Now, we do 50 times more data compared to then and three times the voice, but the ARPU is Rs 150. After adjusted inflation, it should have been Rs 300-plus," says a senior official with a service provider.

One of the primary reasons for this is the cut-throat competition in the sector, especially with the entry of Reliance Jio, which led to rock-bottom tariffs. Even after a 20 per cent increase in tariffs in November 2021, the cheapest tariff plan for Jio stood at just Rs 75 a month, while for Airtel and Vodafone Idea, it was Rs 99 each. Service providers tread cautiously on the issue of tariffs since any incremental increase leads to a massive drop in the number of subscribers. With the advent of 5G and a subsequent effect on speed and latency, data consumption is expected to rise. This could mean more revenues for the service providers. It will be critical for players such as Vodafone Idea, which has been struggling under a huge debt burden, prompting the Centre to come up with a relief package last year that included payment of pending spectrum dues in a staggered manner over 20 years.

IMPLEMENTATION CHALLENGES

The biggest entry barrier for the end-customer would be 5G handsets, which at Rs 15,000 and above are out of reach for many. Most Android devices and iPhones are already introducing 5G-enabled phones in the market. Moreover, only 5-7 per cent of the 800 million smartphones used in the country are 5G-enabled. But minister Vaishnaw claims this will change rapidly. "Close to 20 per cent of the handsets being manufactured in India are 5G-enabled," he says. "As for cost, entry-level 5G handsets have already come in at Rs 15,000. Very soon, it will come down to Rs 10,000, which is a critical point in terms of the cost of handsets. So we expect a big scaling up soon."

Telecom observers expect two kinds of customers. One is those who will continue to use 2G, especially in the villages. This ecosystem will endure for the next five years, they say. The second set will be customers hungry for data content, those who have already scaled up from social messaging on apps like WhatsApp to more data-heavy use, such as video calling. This set of users will seamlessly move into 5G.

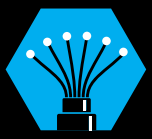
Telecom companies are busy stitching up alliances and sprucing up technology in preparation for the 5G launch. Bharti Airtel has tied up with Tech Mahindra to create and market enterprise-grade digital solutions spanning 5G, private networks and the cloud. It is also roping in equipment providers to negotiate 5G-specific contracts. Reliance Jio is in advanced talks with South Korea's Samsung as a third-party service provider. It is also collaborating with Google Cloud on 5G. Meanwhile, Vodafone Idea is partnering with US-based Ciena to con-

CHALLENGES



At over 700,000, India has the largest number of telecom towers in the world, but it still falls short. Experts say that the proliferation coverage of telecom towers has to go up to nearly twice as much for a smooth rollout of 5G

Deep fiberisation of the network so that towers do not get choked. Fiberisation is only around 30 per cent at present, and needs to increase to 80 per cent for a meaningful rollout of 5G





There are still some 400 million subscribers running on 2G/3G networks and devices, and the industry needs to convince them to move to 4G and beyond

As many as 100 million feature phones are still being sold each year in India, while the sale of smartphones is in the region of 170-180 million. Today, the average price for a 5G-enabled handset is Rs 15,000. For large-scale proliferation of 5G, devices in the Rs 5,000-7,000 price range are required





Massive investments are needed in infrastructure, but some telecom players are struggling with huge debt as well as repayment of spectrum dues

URBAN AREAS WITH **HIGHER TELECOM TOWER DENSITY AND A HIGHER NUMBER OF 5G HANDSETS** WILL BE THE FIRST TO WITNESS THE IMPLEMENTATION